Rank-based pairwise distance matrix

We calculate the rank-based genealogical distances between a pair of sequences, we discussed how we might set the ranks of the unobserved coalescence events on the partial genealogy. These unobserved coalescence events are more ancient than the stopping time τ. We proposed setting the ranks of these more ancient coalescence events to all be n-1 or some large numbers, where n is the number of sequences in the sample. However, we then think this scheme may overemphasize the ancient topology and take focus away from the more recent coalescence events that we do observe on the partial genealogy. This is not what we want. Instead, we want to emphasize the coalescence events that are more recent than τ because these events should be more informative for deleterious rare variants that lead to disease traits.

Therefore, we propose to assign the ranks for the more ancient coalescence events (than time τ) so that all the ranks add to 1+2+...+(n-1) = n(n-1)/2. This should calibrate the genealogical distance between any pair of sequences in different clades at the stopping time to be more similar to their genealogical distance in the full genealogical tree (with all coalescence events observed).

In a nutshell, the idea is to constrain the ranks of the coalescence times to sum as they would in a full genealogy. Let t_i be the time of the ith most recent coalescence event. E.G., t_1 is the time of the 1st most recent coalescence at which there are n-1 lineages, t_2 is the time of the 2nd most recent coalescence at which there are n-2 lineages, etc. Finally, $t_{(n-1)}$ is the time of the final coalescence, at which there is n-(n-1)=1 lineage. In general, at index (or rank) i, there are n-i lineages. Alternatively, letting x=n-i be the number of lineages, the index (or rank) of the corresponding coalescence event is i=n-x. Define the stopping time τ to be the most recent time at which there are n_τ lineages. Then the index (or rank) for the coalescence event corresponding to n_τ is $n - n_\tau$. We want to assign ranks to the unobserved coalescence events in the partial genealogy. These unobserved coalescences are more ancient than the stopping time τ and have ranks greater than $n-n_\tau$. Here we describe how we assign ranks to these coalescence events so that all the ranks add to n(n-1)/2. For the full genealogy, we know that $(n-1)n/2 = rank(t_1) + rank(t_2) + ...rank(t_{n-1}) = 1 + 2 + ... + (n-1)$. The sum of the ranks can be broken down into two smaller sums defined by the stopping time as $\sum_{i=1}^{n-n_\tau} rank(t_i) + \sum_{i=n-n_\tau+1}^{n-1} rank(t_i)$ and we can solve for the sum of the more ancient ranks to get that $\sum_{i=n-n_\tau+1}^{n-1} rank(t_i) = n(n-1)/2 - (n-n_\tau)(n-n_\tau+1)/2$. Therefore, for $i \in \{n-n_\tau+1, ..., n-1\}$, we should set rank($t_i$) to be $[n(n-1)/2 - (n-n_\tau)(n-n_\tau+1)/2]/[n-1-(n-n_\tau+1)+1] = [n(n-1) - (n-n_\tau)(n-n_\tau+1)]/[2(n_\tau-1)]$. This assigns the same rank to all the ancient coalescence events in the partial genealogy, so that the distance between any pair of sequences in different clades at the stopping time is calibrated to the full genealogical tree if all the coalescence events had been observed.